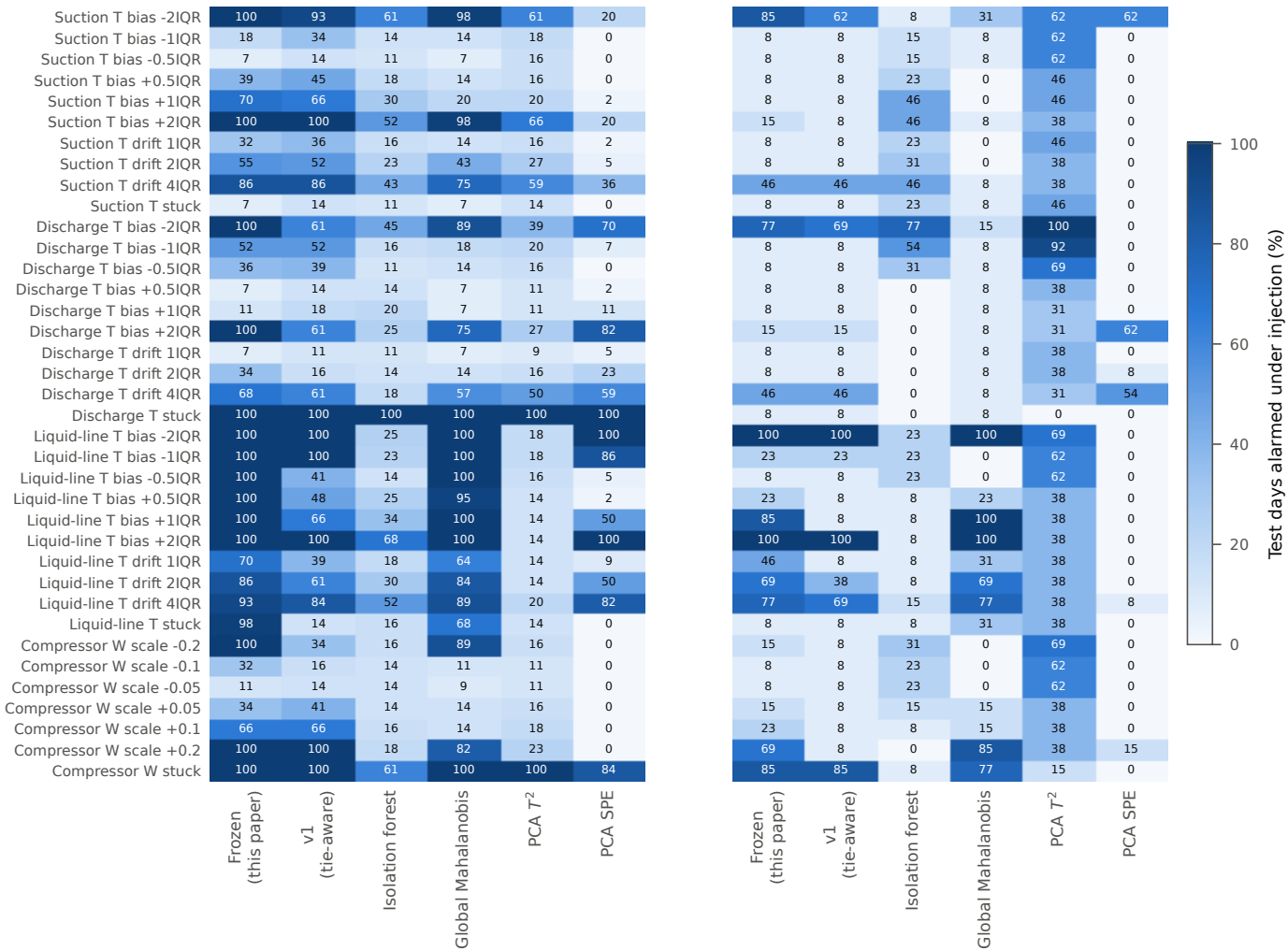
Site 2 (44 normal test days)
Site 1 (13 normal test days)
Suction T bias -2IQR 100 93 61 98 61 20 | 85 62 8 31 62 62
Suction T bias -1IQR 18 34 14 14 18 0 | 8 8 15 8 62 0
Suction T bias -0.5IQR 7 14 11 7 16 0 | 8 8 15 8 62 0
Suction T bias +0.5IQR 39 45 18 14 16 0 | 8 8 23 0 46 0
Suction T bias +1IQR 70 66 30 20 20 2 | 8 8 46 0 46 0
Suction T bias +2IQR 100 100 52 98 66 20 | 15 8 46 8 38 0
Suction T drift 1IQR 32 36 16 14 16 2 | 8 8 23 0 46 0
Suction T drift 2IQR 55 52 23 43 27 5 | 8 8 31 0 38 0
Suction T drift 4IQR 86 86 43 75 59 36 | 46 46 46 8 38 0
Suction T stuck 7 14 11 7 14 0 | 8 8 23 8 46 0
Discharge T bias -2IQR 100 61 45 89 39 70 | 77 69 77 15 100 0
Discharge T bias -1IQR 52 52 16 18 20 7 | 8 8 54 8 92 0
Discharge T bias -0.5IQR 36 39 11 14 16 0 | 8 8 31 8 69 0
Discharge T bias +0.5IQR 7 14 14 7 11 2 | 8 8 0 8 38 0
Discharge T bias +1IQR 11 18 20 7 11 11 | 8 8 0 8 31 0
Discharge T bias +2IQR 100 61 25 75 27 82 | 15 15 0 8 31 62
Discharge T drift 1IQR 7 11 11 7 9 5 | 8 8 0 8 38 0
Discharge T drift 2IQR 34 16 14 14 16 23 | 8 8 0 8 38 8
Discharge T drift 4IQR 68 61 18 57 50 59 | 46 46 0 8 31 54
Discharge T stuck 100 100 100 100 100 100 | 8 8 0 8 0 0
Liquid-line T bias -2IQR 100 100 25 100 18 100 | 100 100 23 100 69 0
Liquid-line T bias -1IQR 100 100 23 100 18 86 | 23 23 23 0 62 0
Liquid-line T bias -0.5IQR 100 41 14 100 16 5 | 8 8 23 0 62 0
Liquid-line T bias +0.5IQR 100 48 25 95 14 2 | 23 8 8 23 38 0
Liquid-line T bias +1IQR 100 66 34 100 14 50 | 85 8 8 100 38 0
Liquid-line T bias +2IQR 100 100 68 100 14 100 | 100 100 8 100 38 0
Liquid-line T drift 1IQR 70 39 18 64 14 9 | 46 8 8 31 38 0
Liquid-line T drift 2IQR 86 61 30 84 14 50 | 69 38 8 69 38 0
Liquid-line T drift 4IQR 93 84 52 89 20 82 | 77 69 15 77 38 8
Liquid-line T stuck 98 14 16 68 14 0 | 8 8 8 31 38 0
Compressor W scale -0.2 100 34 16 89 16 0 | 15 8 31 0 69 0
Compressor W scale -0.1 32 16 14 11 11 0 | 8 8 23 0 62 0
Compressor W scale -0.05 11 14 14 9 11 0 | 8 8 23 0 62 0
Compressor W scale +0.05 34 41 14 14 16 0 | 15 8 15 15 38 0
Compressor W scale +0.1 66 66 16 14 18 0 | 23 8 8 15 38 0
Compressor W scale +0.2 100 100 18 82 23 0 | 69 8 0 85 38 15
Compressor W stuck 100 100 61 100 100 84 | 85 85 8 77 15 0
Frozen (this paper)
v1 (tie-aware)
Isolation forest
Global Mahalanobis
PCA T^2
PCA SPE
Test days alarmed under injection (%)
0
20
40
60
80
100